ALGEBRA 2 · UNIT 2: LINEAR FUNCTIONS & SYSTEMS

Farm Planner

Use linear programming and systems in three variables to run a farm.

CP1

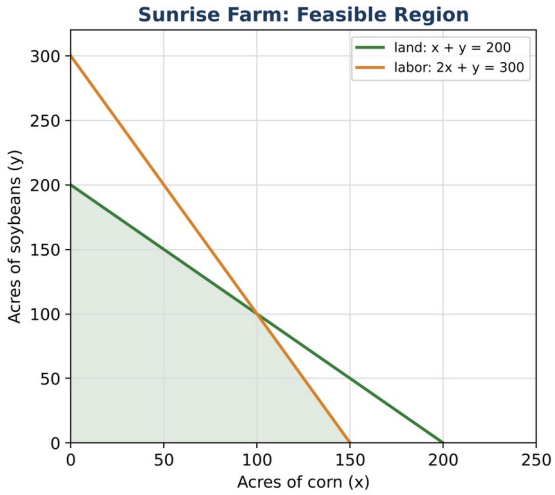
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

Sunrise Farm grows corn and soybeans. Land and labor are limited, and each crop brings a different profit. The farmer wants the plan that earns the most money.

This is linear programming: graph the constraints, find the corners of the feasible region, and test them. Then you will mix feed for the farm's animals with a three-variable system.



Per Acre
Corn: 2 h labor
profit $300
Soybeans: 1 h labor
profit $200
Land ≤ 200 acres
Labor ≤ 300 h

Learning Goals

- I can model constraints with a system of inequalities.
- I can optimize with linear programming.
- I can solve systems of three equations.

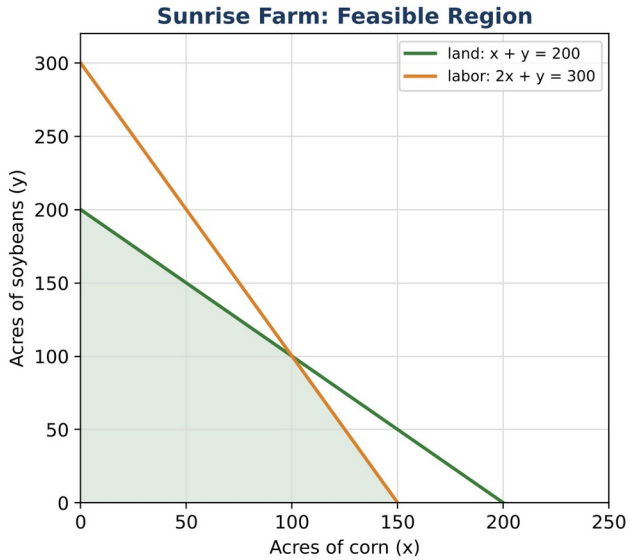
Project Timeline

When	What you will do
Day 1	Constraints
Day 2	Optimize
Day 3	Mix the Feed
Final	Farm Business Plan + reflection

Part 1: Constraints

Day 1 · Use the Per Acre card.

1. Write all constraints and graph the feasible region.



Per Acre

Corn: 2 h labor

profit $300

Soybeans: 1 h labor

profit $200

Land ≤ 200 acres

Labor ≤ 300 h

2. Find all vertices algebraically.

3. Explain why the labor constraint makes some land-only plans impossible.

Part 2: Optimize

Day 2 · Profit P = 300x + 200y.

4. Find the maximum profit and the plan that gives it.

5. If corn profit drops to $150 per acre, what is the new best plan?

6. The farmer adds a rule: at least 40 acres of corn. Add it to the graph. Does the best plan change?

Part 3: Mix the Feed

Day 3 · Feed uses corn (a), soy meal (b), and oats (c), in pounds.

7. A 100-lb batch must have 16.6 lb of protein and cost $15. Protein: corn 9%, soy 44%, oats 12%. Cost per lb: $0.10, $0.30, $0.15. Write the system.

8. Solve the system by elimination and check.

9. Write a business plan with the crop plan and the feed recipe.

Final Product: Farm Business Plan

Your final product must include:

- Graph with all constraints and vertices
- Optimization with a what-if analysis
- Three-variable feed solution with check
- Business plan

Reflection

R1. Why do changes in profit per acre change the best corner?

R2. How is solving a 3 × 3 system like a 2 × 2 system?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Recommendation is the true optimum (or a tie is explained), and the feed system is solved and checked. Goes beyond the minimum with a creative, realistic choice.	Recommendation is the true optimum (or a tie is explained), and the feed system is solved and checked.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Business plan includes the constraints, graph, vertices, optimal plan, and the feed mix solution. Polished, organized, and visually engaging.	Business plan includes the constraints, graph, vertices, optimal plan, and the feed mix solution. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.